

Limit & Infinities

Wednesday, 16 September 2026

11:14 pm

Limits Involving Infinity; Asymptotes of Graphs

Finite Limits as $x \rightarrow \pm\infty$

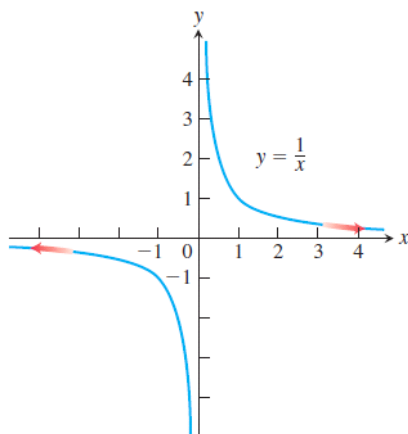


FIGURE 2.49 The graph of $y = 1/x$ approaches 0 as $x \rightarrow \infty$ or $x \rightarrow -\infty$.

THEOREM 12 All the Limit Laws in Theorem 1 are true when we replace $\lim_{x \rightarrow c}$ by $\lim_{x \rightarrow \infty}$ or $\lim_{x \rightarrow -\infty}$. That is, the variable x may approach a finite number c or $\pm\infty$.

EXAMPLE 2 The properties in Theorem 12 are used to calculate limits in the same way as when x approaches a finite number c .

$$\begin{aligned} \text{(a)} \quad \lim_{x \rightarrow \infty} \left(5 + \frac{1}{x} \right) &= \lim_{x \rightarrow \infty} 5 + \lim_{x \rightarrow \infty} \frac{1}{x} && \text{Sum Rule} \\ &= 5 + 0 = 5 && \text{Known limits} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \lim_{x \rightarrow -\infty} \frac{\pi\sqrt{3}}{x^2} &= \lim_{x \rightarrow -\infty} \pi\sqrt{3} \cdot \frac{1}{x} \cdot \frac{1}{x} \\ &= \lim_{x \rightarrow -\infty} \pi\sqrt{3} \cdot \lim_{x \rightarrow -\infty} \frac{1}{x} \cdot \lim_{x \rightarrow -\infty} \frac{1}{x} && \text{Product Rule} \\ &= \pi\sqrt{3} \cdot 0 \cdot 0 = 0 && \text{Known limits} \end{aligned}$$

$$f(x) = 4$$

$$f(x) = 4x$$

$$f(x) = -x^2$$

$$f(x) = e^{-x}$$

$$f(x) = 4(1 - e^{-x})$$

$$f(x) = \cos(x)$$

$$\lim_{x \rightarrow \infty} f(x) = L$$

$$-\infty < L < \infty$$

$$\lim_{x \rightarrow \infty} f(x) = \infty$$

$$\lim_{x \rightarrow \infty} f(x) \text{ DNE}$$

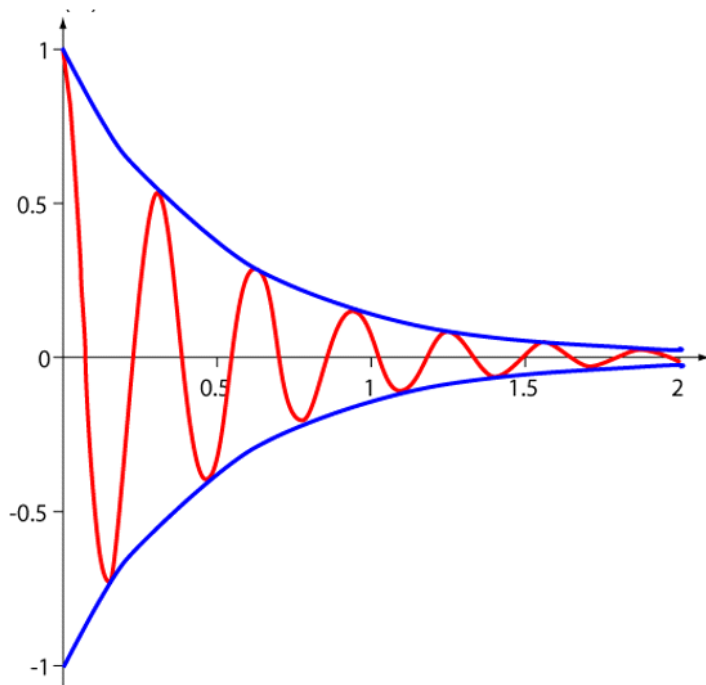
Interesting case...



$$f(x) = e^{-x} \cos(x)$$

What happens to this function as $x \rightarrow \infty$?

- Let's sketch it.
- Clearly the oscillation is dying out exponentially.



$$\lim_{x \rightarrow \infty} f(x) = 0$$

Interesting case...



$$\lim_{x \rightarrow \infty} \left(\frac{x}{x} \right) = 1$$

Hint: plug in larger and larger values of x . Do we keep approaching a fixed value? What is it?

- Q. Why can't we just cancel out the two x 's and then plug in $x = \infty$?
- Because ∞ is not in the domain of any function (it is, in fact, not a number at all).
- In addition, when we cancel the x 's, we implicitly assume that the x takes only those values where division is possible.
- Otherwise we are claiming $\frac{\infty}{\infty} = 1$

Interesting cases...



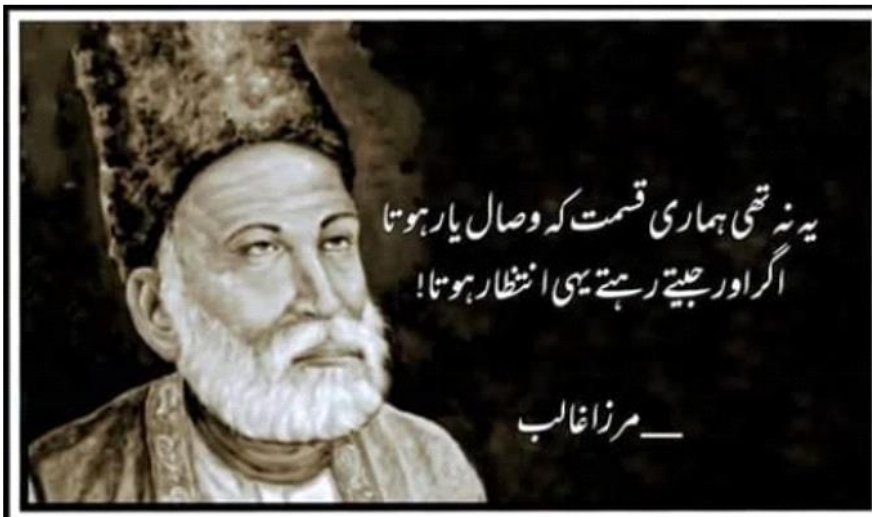
$$\lim_{x \rightarrow \infty} \left(\frac{x^2}{x} \right) = ?$$

Hint: plug in larger and larger values of x . Do we keep approaching a fixed value? Or do we see that the output keeps increasing without bound?

$$\lim_{x \rightarrow \infty} \left(\frac{x}{x^2} \right) = ?$$

Hint: plug in larger and larger values of x . Do we keep approaching a fixed value? What is it?

Limits in Poetry



Let t be time and
wisaal/union be W . Then...

$$\lim_{t \rightarrow \infty} W(t) = 0$$

مریض عشق پہ رحمت خدا کی
مرض بڑھتا گیا جوں جوں دوا کی

Let x be the amount of treatment,
and M be the maraz/sickness. Then...

$$\lim_{x \rightarrow \infty} M(x) = \infty$$

Limits at Infinity of Rational Functions

EXAMPLE 3 These examples illustrate what happens when the degree of the numerator is less than or equal to the degree of the denominator.

$$\begin{aligned} \text{(a)} \quad \lim_{x \rightarrow \infty} \frac{5x^2 + 8x - 3}{3x^2 + 2} &= \lim_{x \rightarrow \infty} \frac{5 + (8/x) - (3/x^2)}{3 + (2/x^2)} && \text{Divide numerator and denominator by } x^2. \\ &= \frac{5 + 0 - 0}{3 + 0} = \frac{5}{3} && \text{See Fig. 2.51.} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \lim_{x \rightarrow -\infty} \frac{11x + 2}{2x^3 - 1} &= \lim_{x \rightarrow -\infty} \frac{(11/x^2) + (2/x^3)}{2 - (1/x^3)} && \text{Divide numerator and denominator by } x^3. \\ &= \frac{0 + 0}{2 - 0} = 0 && \text{See Fig. 2.52.} \end{aligned}$$

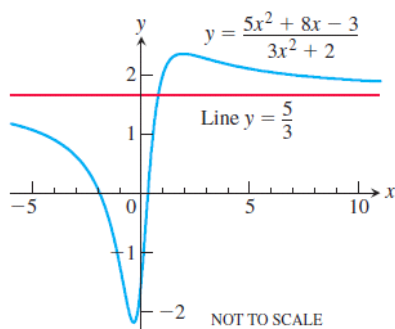


FIGURE 2.51 The graph of the function in Example 3a. The graph approaches the line $y = 5/3$ as $|x|$ increases.

Horizontal Asymptotes

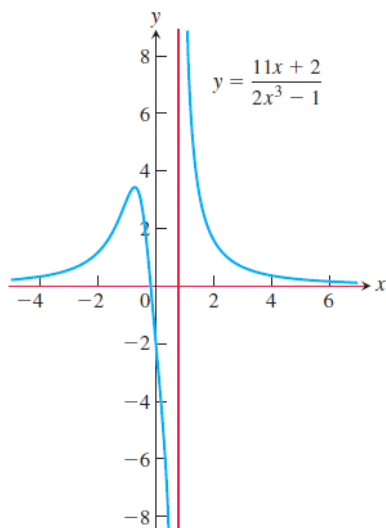


FIGURE 2.52 The graph of the function in Example 3b. The graph approaches the x -axis as $|x|$ increases.

If the distance between the graph of a function and some fixed line approaches zero as a point on the graph moves increasingly far from the origin, we say that the graph approaches the line asymptotically and that the line is an *asymptote* of the graph.

Looking at $f(x) = 1/x$ (see Figure 2.49), we observe that the x -axis is an asymptote of the curve on the right because

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

and on the left because

$$\lim_{x \rightarrow -\infty} \frac{1}{x} = 0.$$

We say that the x -axis is a *horizontal asymptote* of the graph of $f(x) = 1/x$.

DEFINITION A line $y = b$ is a **horizontal asymptote** of the graph of a function $y = f(x)$ if either

$$\lim_{x \rightarrow \infty} f(x) = b \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = b.$$

The graph of the function

$$f(x) = \frac{5x^2 + 8x - 3}{3x^2 + 2}$$

sketched in Figure 2.51 (Example 3a) has the line $y = 5/3$ as a horizontal asymptote on both the right and the left because

$$\lim_{x \rightarrow \infty} f(x) = \frac{5}{3} \quad \text{and} \quad \lim_{x \rightarrow -\infty} f(x) = \frac{5}{3}.$$

EXAMPLE 4 Find the horizontal asymptotes of the graph of

$$f(x) = \frac{x^3 - 2}{|x|^3 + 1}.$$

Solution We calculate the limits as $x \rightarrow \pm\infty$.

$$\text{For } x \geq 0: \lim_{x \rightarrow \infty} \frac{x^3 - 2}{|x|^3 + 1} = \lim_{x \rightarrow \infty} \frac{x^3 - 2}{x^3 + 1} = \lim_{x \rightarrow \infty} \frac{1 - (2/x^3)}{1 + (1/x^3)} = 1.$$

$$\text{For } x < 0: \lim_{x \rightarrow -\infty} \frac{x^3 - 2}{|x|^3 + 1} = \lim_{x \rightarrow -\infty} \frac{x^3 - 2}{(-x)^3 + 1} = \lim_{x \rightarrow -\infty} \frac{1 - (2/x^3)}{-1 + (1/x^3)} = -1.$$

The horizontal asymptotes are $y = -1$ and $y = 1$. The graph is displayed in Figure 2.53. Notice that the graph crosses the horizontal asymptote $y = -1$ for a positive value of x . ■

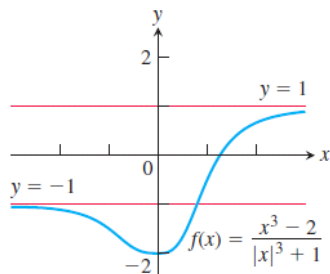


FIGURE 2.53 The graph of the function in Example 4 has two horizontal asymptotes.

EXAMPLE 5 The x -axis (the line $y = 0$) is a horizontal asymptote of the graph of $y = e^x$ because

$$\lim_{x \rightarrow -\infty} e^x = 0.$$

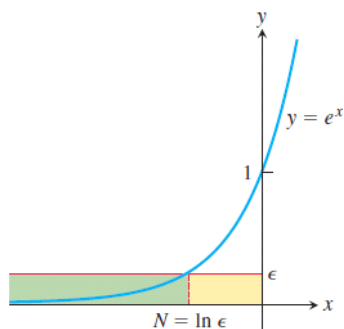


FIGURE 2.54 The graph of $y = e^x$ approaches the x -axis as $x \rightarrow -\infty$ (Example 5).

EXAMPLE 6 Find (a) $\lim_{x \rightarrow \infty} \sin(1/x)$ and (b) $\lim_{x \rightarrow \pm\infty} x \sin(1/x)$.

Solution

- (a) We introduce the new variable $t = 1/x$. From Example 1, we know that $t \rightarrow 0^+$ as $x \rightarrow \infty$ (see Figure 2.49). Therefore,

$$\lim_{x \rightarrow \infty} \sin \frac{1}{x} = \lim_{t \rightarrow 0^+} \sin t = 0.$$

(b) We calculate the limits as $x \rightarrow \infty$ and $x \rightarrow -\infty$:

$$\lim_{x \rightarrow \infty} x \sin \frac{1}{x} = \lim_{t \rightarrow 0^+} \frac{\sin t}{t} = 1 \quad \text{and} \quad \lim_{x \rightarrow -\infty} x \sin \frac{1}{x} = \lim_{t \rightarrow 0^-} \frac{\sin t}{t} = 1.$$

The graph is shown in Figure 2.55, and we see that the line $y = 1$ is a horizontal asymptote.

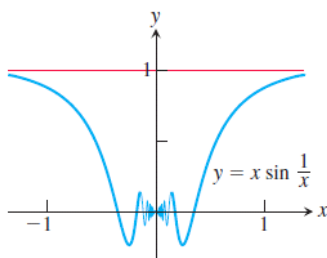


FIGURE 2.55 The line $y = 1$ is a horizontal asymptote of the function graphed here (Example 6b).

EXAMPLE 7 Find $\lim_{x \rightarrow 0^-} e^{1/x}$.

Solution We let $t = 1/x$. From Figure 2.49, we can see that $t \rightarrow -\infty$ as $x \rightarrow 0^-$. (We make this idea more precise further on.) Therefore,

$$\lim_{x \rightarrow 0^-} e^{1/x} = \lim_{t \rightarrow -\infty} e^t = 0 \quad \text{Example 5}$$

(Figure 2.56).

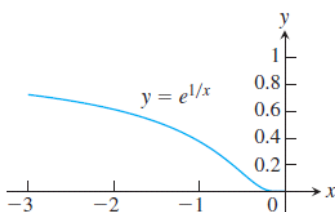


FIGURE 2.56 The graph of $y = e^{1/x}$ for $x < 0$ shows $\lim_{x \rightarrow 0^-} e^{1/x} = 0$ (Example 7).

EXAMPLE 8 Using the Sandwich Theorem, find the horizontal asymptote of the curve

$$y = 2 + \frac{\sin x}{x}.$$

Solution We are interested in the behavior as $x \rightarrow \pm\infty$. Since

$$0 \leq \left| \frac{\sin x}{x} \right| \leq \left| \frac{1}{x} \right|$$

and $\lim_{x \rightarrow \pm\infty} |1/x| = 0$, we have $\lim_{x \rightarrow \pm\infty} (\sin x)/x = 0$ by the Sandwich Theorem. Hence,

$$\lim_{x \rightarrow \pm\infty} \left(2 + \frac{\sin x}{x} \right) = 2 + 0 = 2,$$

and the line $y = 2$ is a horizontal asymptote of the curve on both left and right (Figure 2.57).

This example illustrates that a curve may cross one of its horizontal asymptotes many times.

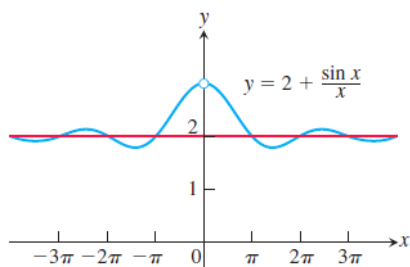


FIGURE 2.57 A curve may cross one of its asymptotes infinitely often (Example 8).

EXAMPLE 9 Find $\lim_{x \rightarrow \infty} (x - \sqrt{x^2 + 16})$.

$$\begin{aligned} \lim_{x \rightarrow \infty} (x - \sqrt{x^2 + 16}) &= \lim_{x \rightarrow \infty} (x - \sqrt{x^2 + 16}) \frac{x + \sqrt{x^2 + 16}}{x + \sqrt{x^2 + 16}} \\ &= \lim_{x \rightarrow \infty} \frac{x^2 - (x^2 + 16)}{x + \sqrt{x^2 + 16}} = \lim_{x \rightarrow \infty} \frac{-16}{x + \sqrt{x^2 + 16}}. \end{aligned}$$

$$\lim_{x \rightarrow \infty} \frac{-16}{x + \sqrt{x^2 + 16}} = \lim_{x \rightarrow \infty} \frac{-\frac{16}{x}}{1 + \sqrt{\frac{x^2}{x^2} + \frac{16}{x^2}}} = \frac{0}{1 + \sqrt{1 + 0}} = 0.$$

Oblique Asymptotes

If the degree of the numerator of a rational function is 1 greater than the degree of the denominator, the graph has an **oblique** or **slant line asymptote**. We find an equation for the asymptote by dividing numerator by denominator to express f as a linear function plus a remainder that goes to zero as $x \rightarrow \pm \infty$.

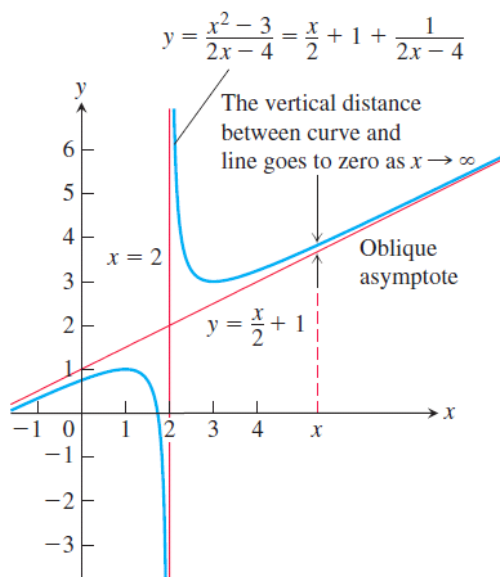


FIGURE 2.58 The graph of the function in Example 10 has an oblique asymptote.

EXAMPLE 10 Find the oblique asymptote of the graph of

$$f(x) = \frac{x^2 - 3}{2x - 4}$$

in Figure 2.58.

Solution We are interested in the behavior as $x \rightarrow \pm\infty$. We divide $(x^2 - 3)$ into $(2x - 4)$:

$$\begin{array}{r} \frac{x}{2} + 1 \\ 2x - 4 \overline{) x^2 - 3} \\ \underline{x^2 - 2x} \\ 2x - 3 \\ \underline{2x - 4} \\ 1 \end{array}$$

This tells us that

$$f(x) = \frac{x^2 - 3}{2x - 4} = \underbrace{\left(\frac{x}{2} + 1\right)}_{\text{linear } g(x)} + \underbrace{\left(\frac{1}{2x - 4}\right)}_{\text{remainder}}.$$

Infinite Limits

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{1}{x} = \infty.$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty.$$

EXAMPLE 11 Find $\lim_{x \rightarrow 1^+} \frac{1}{x - 1}$ and $\lim_{x \rightarrow 1^-} \frac{1}{x - 1}$.

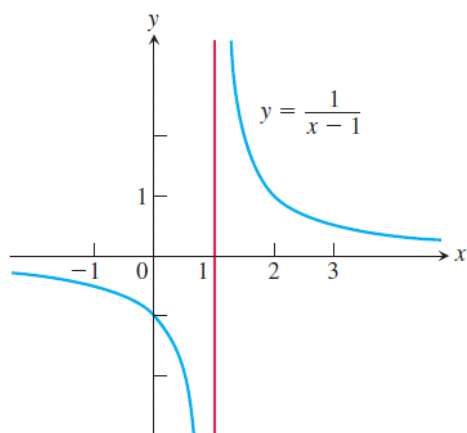


FIGURE 2.60 Near $x = 1$, the function $y = 1/(x - 1)$ behaves the way the function $y = 1/x$ behaves near $x = 0$. Its graph is the graph of $y = 1/x$ shifted 1 unit to the right (Example 11).

EXAMPLE 12 Discuss the behavior of

$$f(x) = \frac{1}{x^2} \quad \text{as} \quad x \rightarrow 0.$$

Solution As x approaches zero from either side, the values of $1/x^2$ are positive and become arbitrarily large (Figure 2.61). This means that

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{1}{x^2} = \infty.$$

The function $y = 1/x$ shows no consistent behavior as $x \rightarrow 0$. We have $1/x \rightarrow \infty$ if $x \rightarrow 0^+$, but $1/x \rightarrow -\infty$ if $x \rightarrow 0^-$. All we can say about $\lim_{x \rightarrow 0} (1/x)$ is that it does not exist. The function $y = 1/x^2$ is different. Its values approach infinity as x approaches zero from either side, so we can say that $\lim_{x \rightarrow 0} (1/x^2) = \infty$. ■

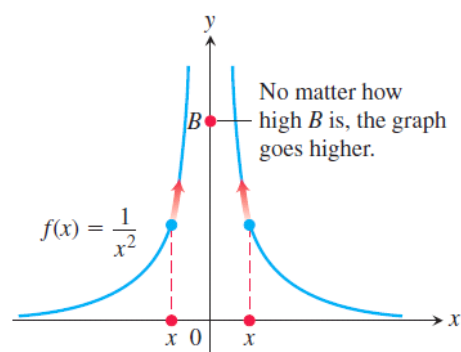


FIGURE 2.61 The graph of $f(x)$ in Example 12 approaches infinity as $x \rightarrow 0$.

EXAMPLE 13 These examples illustrate that rational functions can behave in various ways near zeros of the denominator.

$$(a) \lim_{x \rightarrow 2} \frac{(x-2)^2}{x^2-4} = \lim_{x \rightarrow 2} \frac{(x-2)^2}{(x-2)(x+2)} = \lim_{x \rightarrow 2} \frac{x-2}{x+2} = 0$$

$$(b) \lim_{x \rightarrow 2} \frac{x-2}{x^2-4} = \lim_{x \rightarrow 2} \frac{x-2}{(x-2)(x+2)} = \lim_{x \rightarrow 2} \frac{1}{x+2} = \frac{1}{4}$$

$$(c) \lim_{x \rightarrow 2^+} \frac{x-3}{x^2-4} = \lim_{x \rightarrow 2^+} \frac{x-3}{(x-2)(x+2)} = -\infty$$

The values are negative for $x > 2$, x near 2.

$$(d) \lim_{x \rightarrow 2^-} \frac{x-3}{x^2-4} = \lim_{x \rightarrow 2^-} \frac{x-3}{(x-2)(x+2)} = \infty$$

The values are positive for $x < 2$, x near 2.

$$(e) \lim_{x \rightarrow 2} \frac{x-3}{x^2-4} = \lim_{x \rightarrow 2} \frac{x-3}{(x-2)(x+2)} \text{ does not exist.}$$

See parts (c) and (d).

$$(f) \lim_{x \rightarrow 2} \frac{2-x}{(x-2)^3} = \lim_{x \rightarrow 2} \frac{-(x-2)}{(x-2)^3} = \lim_{x \rightarrow 2} \frac{-1}{(x-2)^2} = -\infty$$

In parts (a) and (b) the effect of the zero in the denominator at $x = 2$ is canceled because the numerator is zero there also. Thus a finite limit exists. This is not true in part (f), where cancellation still leaves a zero factor in the denominator. ■

EXAMPLE 14 Find $\lim_{x \rightarrow -\infty} \frac{2x^5 - 6x^4 + 1}{3x^2 + x - 7}$.

Solution We are asked to find the limit of a rational function as $x \rightarrow -\infty$, so we divide the numerator and denominator by x^2 , the highest power of x in the denominator:

$$\begin{aligned} \lim_{x \rightarrow -\infty} \frac{2x^5 - 6x^4 + 1}{3x^2 + x - 7} &= \lim_{x \rightarrow -\infty} \frac{2x^3 - 6x^2 + x^{-2}}{3 + x^{-1} - 7x^{-2}} \\ &= \lim_{x \rightarrow -\infty} \frac{2x^2(x-3) + x^{-2}}{3 + x^{-1} - 7x^{-2}} \\ &= -\infty, \quad x^{-n} \rightarrow 0, \quad x-3 \rightarrow -\infty \end{aligned}$$

because the numerator tends to $-\infty$ while the denominator approaches 3 as $x \rightarrow -\infty$. ■

Vertical Asymptotes

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty \quad \text{and} \quad \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty.$$

DEFINITION A line $x = a$ is a **vertical asymptote** of the graph of a function $y = f(x)$ if either

$$\lim_{x \rightarrow a^+} f(x) = \pm\infty \quad \text{or} \quad \lim_{x \rightarrow a^-} f(x) = \pm\infty.$$

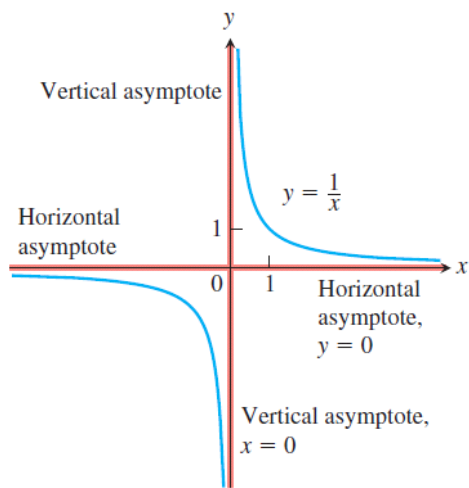


FIGURE 2.64 The coordinate axes are asymptotes of both branches of the hyperbola $y = 1/x$.

EXAMPLE 16 Find the horizontal and vertical asymptotes of the curve

$$y = \frac{x + 3}{x + 2}.$$

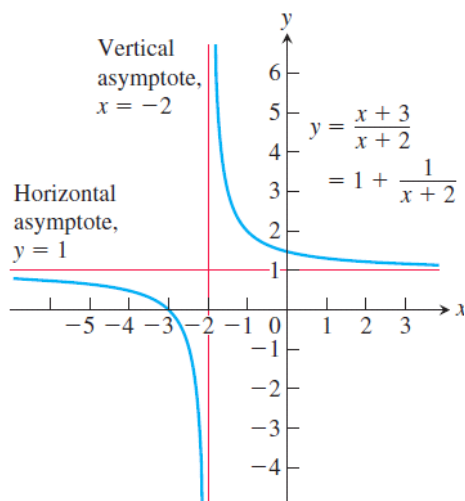


FIGURE 2.65 The lines $y = 1$ and $x = -2$ are asymptotes of the curve in Example 16.

EXAMPLE 17 Find the horizontal and vertical asymptotes of the graph of

$$f(x) = -\frac{8}{x^2 - 4}.$$

